Sara Ghanbari

M.Sc. Computer Science Student | Machine Learning & Applied AI
linkedin.com/in/sara-ghanbari | saraghanbari.sg@gmail.com | +49 15214956332 | Saarbrücken, Germany

Summary

M.Sc. Computer Science student specializing in machine learning and applied AI, with experience developing deep-learning systems for anomaly detection and biometric identification. Skilled in model development, data preparation, evaluation, and applied AI integration, with research publications in biometric recognition.

Education

2024 - Present **M.Sc. Computer Science**, Saarland University, Germany
Relevant Coursework: Machine Learning, Neural Networks, Reinforcement Learning, Deep RL, Graph Neural Networks, Trustworthy ML, Data Science, HCI.

2018 - 2023 **B.Sc. Electrical Engineering**, University of Tehran, Iran; Major: Control Systems; GPA: 17.21/20
Relevant Coursework: Neural Networks, Artificial Intelligence, Advanced Programming.

Experience

AI Engineer, Setare Rahnama Sanat Company 2023 - 2024
- Developed a computer-vision anomaly-detection web application that converted raw security-camera footage into real-time monitoring outputs for identifying abnormal industrial activities.
- Designed a video-data preparation workflow to clean, structure, and organize raw footage into supervised datasets for model training, validation, and systematic evaluation.
- Integrated YOLOv9-based object detection models into the application interface, enabling visual predictions, automated alerts, and end-to-end anomaly-monitoring workflows in real-world industrial environments.

Teaching Assistant, University of Tehran 2022
- Served as Teaching Assistant for a graduate-level Neural Networks course, supporting project design, grading, and feedback for 150+ students.
- Graded assignments and exams for Modern Control, Linear Algebra, and Operational Research courses, assisted with classes of 75–90+ students.

Research Assistant, Human and Robot Interaction Laboratory 2020 - 2022
- Developed deep-learning approaches for hand biometric recognition, supporting user identification in human-robot interaction systems.
- Built curated hand-geometry datasets for training, validation, and benchmarking, enabling more reliable evaluation of biometric-recognition models.
- Designed and evaluated a custom DNN for hand biometric recognition, contributing to prototype-device validation and two published biometric-identification papers.

Publications

- "User Identification Based on Hand Geometrical Biometrics Using MediaPipe," ICEE 2022.
- "Deep-HFID: Deep Neural Network-Based Hands and Face Biometric Identification System Using Metric Learning," ICRoM 2022.

Selected Academic Projects

Hand Biometrics Feature Selection – Applied separation-index analysis to evaluate DNN representations and identify discriminative biometric features.
CNN Representation Analysis with VGG16 – Analyzed feature separability across VGG16 layers to assess deep representation quality.
Face Recognition on LFW – Built a biometric verification pipeline using face embeddings, metric learning, and similarity scoring; evaluated identity-matching performance on the LFW benchmark.
DCGAN for Emoji Dataset – Trained a convolutional GAN for emoji image generation and evaluated generated sample quality during adversarial training.
Instrument Note Detection – Compared RNN, LSTM, and GRU models for sequence-based musical note classification.

Skills

Machine Learning:	Deep Learning, Model Evaluation, Data Preprocessing, CNNs, RNNs, GANs, Reinforcement Learning, Metric Learning, GNNs, Computer Vision
Frameworks:	PyTorch, TensorFlow, Scikit-learn
Programming:	Python, C/C++, MATLAB, JavaScript, HTML/CSS, React
Data / Tools:	NumPy, Pandas, SciPy, Matplotlib, Docker, Git, Linux
Languages:	Persian: Native, English: Professional, German: Basic